

# Nonverbal Communication of Confidence in Soccer Referees: An Experimental Test of Darwin's Leakage Hypothesis

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The goal of the present paper was to investigate whether soccer referees' nonverbal behavior (NVB) differed based on the difficulty of their decisions and whether perceivers could detect these systematic variations. On the one hand, communicating confidence via NVB is emphasized in referee training. On the other hand, it seems feasible from a theoretical point of view that particularly following relatively difficult decisions referees have problems controlling their NVB. We conducted three experiments to investigate this question. Experiment 1 ( $N = 40$ ) and Experiment 2 ( $N = 60$ ) provided evidence that perceivers regard referees' NVB as less confident following ambiguous decisions as compared with following unambiguous decisions. Experiment 3 ( $N = 58$ ) suggested that perceivers were more likely to debate with the referee when referees nonverbally communicated less confidence. We discuss consequences for referee training.

**Keywords:** sport, thin slices, confidence, leakage

It's like the old swan effect isn't it, you're paddling like crazy underneath but you're nice and calm and I think the best officials in the world regardless of what sport it is that they've got the swan effect, they can sell their decision, they're cool, they're calm under pressure and that's how we relate corporate theatre as being able to sell the decision even if you're not quite in the right position or you may have perhaps perceivably missed something happening previously (Cunningham, Mellick, Mascarenhas, & Fleming, 2012, p 28).

Both the Fédération Internationale de Football Association (FIFA) and the Union of European Football Associations (UEFA) emphasize the importance of the "right body language" of soccer referees in communicating their decisions. UEFA offers special seminars to referees (UEFA, 2016) to "enhance their performance in the pressure-cooker atmosphere of a major match." In their official publication on the laws of the game FIFA states that "body language is a tool that the referee uses to: help him control the match; and show authority and self-control" (FIFA, 2015, p. 84). These suggestions rest on the premise that referees can deliberately use their

body language to facilitate their performance and communication in leading a match. However, a large body of evidence (see Matsumoto, Frank, & Hwang, 2013, for a review) suggests that body language (from now on referred to as nonverbal behavior; NVB) is under both conscious, deliberate control and unconscious, autonomous control. Of particular importance to the present research, evidence shows that contextual influences like pressure or fatigue tip the balance between unconscious and conscious control of NVB toward unconscious control (e.g., see Matsumoto et al., 2013, for a recent review). Hence, soccer referees might be able to use their NVB to present themselves favorably when relaxed and rested—for example, during a seminar teaching nonverbal communication (UEFA, 2016)—but might have more trouble controlling their NVB during actual competition when thousands of people are watching and evaluating them.

According to evolutionary accounts of nonverbal expressions, humans have evolved to be especially well equipped at communicating important internal states nonverbally in terms of displaying emotional nonverbal expressions and interpreting these (Darwin, 1872/1998; Ekman, 1992; Shariff & Tracy, 2011) or communicating social intentions (Fridlund, 1994). Contemporary theorizing suggests that the communicative function of nonverbal expressions is of vital importance for group-living primates to quickly and efficiently communicate social information in complex societies (see Matsumoto et al., 2013; Shariff & Tracy, 2011, for reviews). Therefore, it has been suggested that humans have evolved the ability both to automatically display NVB and to automatically

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interpret and adequately respond to these NVBs (Ekman, 1992). A line of research providing evidence for the adaptive importance of nonverbal communication—which has been termed the *thin slices* approach—demonstrates that people are highly accurate at predicting various outcomes from short observations of NVB (Ambady, Bernieri, & Richeson, 2000; Ambady & Rosenthal, 1992; Furley & Schweizer, 2014a, 2014b, 2016).

Given the assumed evolutionary utility of communicating internal states nonverbally, it is not surprising that both FIFA and UEFA emphasize the importance of nonverbal communication and try to educate their referees in this communication facet. One of the most important aspects of referees' nonverbal communication is conveying confidence in their decision: "Part of the communication begins in the way you actually blow your whistle and the sounds you're making, and confidence with which you do that" (Cunningham, Simmons, Mascarenhas, & Redhead, 2014, p. 174). Most research on confidence in sport has used Bandura's (1997) self-efficacy framework as underlying theory. However, as this theory does not explicitly cover how nonverbal expressions might show after a behavioral response has occurred, we borrowed a broad definition of confidence from the nonverbal communication literature in which it is defined as a person's belief that a chosen course of action is the best in a given situation (Knapp, Hall, & Horgan, 2013). Importantly, this has been suggested to show in a person's NVB (e.g., Calero, 2005; Knapp et al., 2013). This definition highlights that feeling and portraying confidence is highly sensitive to situational fluctuations (Vealey & Chase, 2008). Professional referees have been shown to be motivated to project impressions of confidence as this helps them to control the game (Cunningham et al., 2012; Thatcher, 2005). In this respect, the utility of body language, facial expressions, and other NVBs was found to be of utmost importance in an interview study among professional referees (Cunningham et al., 2014). Although there was consensus among interviewees about the importance of body language in creating favorable impressions and conveying credibility and authority, referees also agreed that this was a difficult skill to teach (Cunningham et al., 2014).

Differentiating "true" expressions from "staged expressions" has been a topic of major interest within the field of nonverbal communication and has been significantly driven by the work from Paul Ekman (e.g., Ekman, 2003). This field of study is relevant for the present research as it has revealed that there are cues distinguishing expressions coinciding with real feelings and staged expressions (see Ekman, 2003, for a review). These findings rest on Charles Darwin's (1872/1998) theory that muscles that are difficult to voluntarily activate might escape efforts to inhibit or mask expressions, revealing true feelings:

When movements, associated through habit with certain states of the mind, are partially repressed

by the will, the strictly involuntary muscles, as well as those which are least under the separate control of the will, are liable still to act; and their action is often highly expressive. (p. 54)

If the theorizing of Darwin and Ekman is correct, then it seems plausible that referees trying to act confident in their decisions should display *leakage* in their NVB when communicating ambiguous decisions as opposed to unambiguous decisions. We tested this hypothesis in the present research.

## The Present Research

The goal of the present research was to investigate how well professional soccer referees were able to reach their goal of projecting confidence in their decisions, even when communicating relatively difficult decisions. A large body of evidence (see Bogdan, 1997, for a review) demonstrates that internal states can "leak" through via nonverbal expressions, even if people are trying to mask their true feelings: "It appears that social communication has been chosen by natural selection to be of greater survival value than disguising our intentions and feelings, so much so that we even have ways of unintentionally 'outing' ourselves to others" (Cozolino, 2006, p. 24). Furthermore, actively disguising one's NVB requires cognitive resources—that is, working memory capacity (WMC; see Frank & Svetieva, 2013, for a review). As difficult decisions should require more WMC than easy decisions (Furley & Memmert, 2012), and as WMC is a limited resource, dual-process models of behavior would predict that particularly after difficult decisions referees should fail to control their NVB (Furley, Schweizer, & Bertrams, 2015).

Hence, we hypothesized that referees' NVB would also show this kind of leakage or, more specifically, that referees' in-game NVB would be interpreted as less confident when communicating ambiguous in contrast to unambiguous decisions. A decision was classified as unambiguous if different referees had decided unanimously on the respective scenes in a preliminary study and as ambiguous if referees made diverging decisions in the preliminary study.

## Experiment 1: Confidence and NVB of Soccer Referees

### Method

**Participants.** Forty college students (studying sport science and physical education) took part in the study (20 women and 20 men;  $M_{\text{age}} = 24.4$ ,  $SD = 4.2$ ). Eleven reported having played soccer in an organized context at some point in their life. On average the sample reported watching 2.3 hr of soccer per week ( $SD = 2.5$ ). Written informed consent was obtained from every participant before commencing the experiment.

**Stimuli.** We selected video footage from a convenience sample of 30 games of the highest men's soccer league in Germany (Bundesliga season 2014–2015). A research assistant selected all referee decisions during these games. The research assistant asked to sample the videos was blind to the hypothesis. In a second step all scenes were excluded in which the communication of the decision was not clearly visible in the video footage. At the end of this process we had 130 video scenes of fouls, yellow cards, and red cards. The rationale for only selecting fouls, yellow cards, and red cards, and not “no-call situations,” was that the game was paused after these decisions and the footage showed more of the referees' NVB. These scenes were cut with video editing software so that the videos either only showed the foul situation or only showed the communication of the decision. Subsequently we asked three semiprofessional soccer referees ( $M_{\text{age}} = 28.3$  years; second highest national refereeing license) to independently state what their decision would be for every video and how confident they were in their decision on a scale of 1 (*not at all confident*) to 11 (*highly confident*). Based on these ratings we formed two experimental stimulus categories: (1) unambiguous scenes, containing 25 videos for which all three referees had made the same decision (mean confidence ratings = 10.2;  $SD = 0.46$ ), and (2) ambiguous scenes for which the referees had made diverging decisions with lower confidence ratings (mean confidence ratings = 7.4;  $SD = 0.64$ ). Within the ambiguous category referees only gave three different decisions for three scenes whereas a 2:1 ratio (i.e., two same decisions) was evident for 22 scenes. An independent  $t$  test revealed a significant difference between the mean confidence ratings (of the three referees) for the 25 ambiguous and 25 unambiguous scenes ( $p < .001$ ). A further study involving five different semiprofessional referees confirmed the significant differences ( $p < .001$ ) in confidence ratings between the 25 respective videos in the ambiguous and unambiguous scenes. The rationale for replicating the initial assessment with different referees was to make sure that our criterion variable of ambiguous versus unambiguous situations was empirically solid and would therefore allow us to infer that referees' decisions were relatively difficult in the ambiguous scenes and relatively easy in the unambiguous scenes.

These 50 scenes served as the experimental stimuli in the present research. The videos had a mean duration of 8.1 s ( $SD = 4.6$ ;  $Mdn = 6.7$ ) and were cut to ensure that only the communication of the decision was visible to observers and not the foul scene before the decision. All scenes started with the camera on the referee and with his blowing the whistle, gesturing, or showing the respective card. All videos came to an end immediately after communication of the decision, for example, the first frame after lowering the hand that was showing the card. There was no significant difference in duration between ambiguous and unambiguous scenes ( $p = .19$ ). All videos were presented without sound.

**Experimental Manipulation (Within Participants).** The experimental factor was the category of the videos: ambiguous versus unambiguous. The sample size was planned in advance to be large enough to detect small-to-medium effects with a power of .8.

**Measure.** Perceivers rated the short video scenes on an 11-point digital semantic differential scale after every video. They were asked whether they thought that the displayed referee was *not at all confident* or *very confident* in his decision. We decided on a single-item measure of confidence (see Tenenbaum, Kamata, & Hayashi, 2007, for the utility of using a one-item measure of confidence) as untrained observers have been reported to have an intuitive understanding of this internal state and can therefore make reliable judgments (e.g., Ambady & Rosenthal, 1993) as compared with more complex inferences (Harrigan, Rosenthal, & Scherer, 2008). To give their ratings, perceivers moved a mouse cursor from the middle of the scale toward either pole of the scale and logged in their rating by clicking the left mouse button. The software converted the ratings into a value between 0, reflecting the left pole of the scale with the label *not at all confident*, and 100, reflecting the right pole of the scale with the label *very confident*. The used scale was continuous, ranging from 0 to 100, and was visually presented as 11 points (0–10) to assist participants in providing a clear indication of their ratings.

**Procedure.** Before commencing the experiment, perceivers filled out a questionnaire gathering demographic data. Every perceiver was tested individually on a standard 17-in. (about 43.2-cm) notebook placed approximately 60 cm away from the perceivers. E-Prime 2.0 Professional (Psychological Software Tools, Sharpsburg, PA) was used to present the stimuli and collect the judgments. All videos were presented silently to ensure that ratings were based on NVB and not, for example, crowd noise. For every perceiver the software randomly presented all 50 videos from both the ambiguous and unambiguous categories. After every video clip perceivers had to give their rating by clicking the left mouse button on the scale described above. Perceivers were instructed that they had to estimate how confident they thought the presented referee was in his communicated decision based on the video footage by moving a mouse cursor to either pole of the semantic differential scale. They were further instructed to answer as accurately as possible, while speed was not emphasized. After completing the testing procedure, participants were informed of the purpose of the experiment.

## Results

The mean perceived confidence rating was 63.35 ( $SD = 11.80$ ) for the ambiguous scenes and 66.23 ( $SD = 11.46$ ) for the unambiguous scenes. A paired-samples  $t$  test revealed a significant difference,  $t(39) = 2.442$ ,  $p = .01$ , one-tailed,  $d = 0.39$ , between the ratings for the unambiguous and the ambiguous scenes, indicating that

perceivers rated the referee as more confident when communicating unambiguous situations as compared with ambiguous situations. Neither soccer experience nor the amount of soccer participants watched significantly influenced the pattern of results (i.e., these variables were not correlated with the difference between confidence ratings for the ambiguous and unambiguous scenes).

## Discussion

Experiment 1 confirmed the leakage hypothesis by showing that although referees are motivated to project confidence, their NVB seems to leak a lack of confidence when communicating relatively difficult decisions. However, the results from Experiment 1 allow an alternative explanation. As the stimulus material also showed reactions of players while communicating the decision, it remains feasible that ratings were not influenced by the NVB of the referee but were instead based on the reactions of players to the referee's decision. For example they might have argued more with the referee in the ambiguous scenes. To rule out this explanation, we digitally modified the video stimuli by occluding the context of the videos with a black mask and cutting videos to only show the referee.

## Experiment 2: Confidence and NVB of Soccer Referees With Contextual Information Occluded

### Method

**Participants.** Sixty different college students (studying sport science and physical education) took part in the study (32 women and 28 men;  $M_{\text{age}} = 22.7$ ,  $SD = 2.3$ ). Eighteen participants reported having played football at some time in their life. On average the sample reported watching 74 min of soccer per week ( $SD = 103$ ). Written informed consent was obtained from every participant before commencing the experiment.

**Stimuli.** The video footage of Experiment 1 served as basis for the stimuli of Experiment 2. We used Adobe After Effects CS 4 to digitally occlude the video scenes so that only the NVB of the referee was visible. In some instances this was not entirely possible as the referee overlapped with other players in the video footage. Sections that showed players discussing with the referee were cut from the footage. The new videos had a mean duration of 3.6 s ( $SD = 2.5$ ;  $Mdn = 3.0$ ) and were cut to ensure that only the communication of the decision was visible. There was no difference in duration between ambiguous and unambiguous scenes ( $p = .51$ ). All videos were presented without sound.

**Measure and Procedure.** The measure and procedure were identical to those in Experiment 1, with the only exception being that the stimulus material was digitally modified. Given the ongoing discussion about adequate sample sizes (Fraleigh & Vazire, 2014; Schweizer & Furley,

2016), we calculated our sample size to be able to detect small effects with a power of .8, as these effects can be considered typical in psychological research.

## Results

The mean perceived confidence rating was 57.57 ( $SD = 10.08$ ) for the ambiguous scenes and 62.71 ( $SD = 10.69$ ) for the unambiguous scenes. A paired-samples  $t$  test revealed a significant difference,  $t(59) = 5.161$ ,  $p = .001$ , one-tailed,  $d = 0.67$ , between the ratings for the unambiguous and the ambiguous scenes, indicating that perceivers rated the referee as more confident when communicating unambiguous situations as compared with ambiguous situations. Neither soccer experience nor the amount of soccer participants watched significantly influenced the pattern of results (i.e., these variables were not correlated with the difference between confidence ratings for the ambiguous and unambiguous scenes).

## Discussion

Again, Experiment 2 confirmed the leakage hypothesis. Of particular importance, the effect increased from  $d = 0.39$  in Experiment 1 to  $d = 0.67$  in Experiment 2, indicating that it was indeed the NVB of the referees and not the behavior of players influencing the confidence ratings of the perceivers. Hence, the NVB of the referee seems to unintentionally leak information depending on how ambiguous the situation is that the referee is communicating. In this respect the body language of a referee might not always be a tool that a referee can use to help him control the match as stated by FIFA but instead has the potential of unintentionally "outing" him to others (Cozolino, 2006) and in turn causing him to lose control of the match. In Experiment 3 we attempted to test this idea in a hypothetical scenario.

## Experiment 3: Does the Referee's NVB Influence Willingness to Debate the Referee's Decision?

Experiment 3 rests on the premise that NVB influences social interactions (e.g., Matsumoto et al., 2013). Hence, we hypothesized on the basis of Experiments 1 and 2 that athletes would be receptive toward the nonverbal leakage of a lack of confidence in a referee's behavior and that this in turn might influence their own behavior. To test this hypothesis, we had participants in Experiment 3 imagine a hypothetical scenario and asked them how they were likely to respond in this scenario. Contemporary psychological theorizing emphasizes the importance of the unique human capacity for mentally simulating future social interactions in the service of guiding behavior (Gilbert & Wilson, 2007; Schacter, Addis, & Buckner, 2007; Seligman, Railton, Baumeister, & Sripada, 2016). This prospective ability has led to important empirical insights in explaining behavior (see Seligman et al., 2016, for a

recent review), including in the field of NVB in sports (Furley, Dicks, Stendtkke, & Memmert, 2012; Furley & Schweizer, 2014a).

Therefore, participants were asked to take over the role of a soccer player involved in a match. Further, they were instructed that they would be shown situations of referees who had just communicated a decision to them with which they did not agree and indicate how likely it was that they would challenge this decision by arguing with the referee. We hypothesized that participants would be more likely to argue with referees communicating decisions in ambiguous scenes as opposed to unambiguous scenes.

## Method

**Participants.** Fifty-eight different college students (studying sport science and physical education) took part in the study (30 women and 28 men;  $M_{age} = 23.5$ ,  $SD = 2.8$ ). Twenty participants reported having played football at some time in their life. On average the sample reported watching 54 min of soccer per week ( $SD = 72$ ). Written informed consent was obtained from every participant before commencing the experiment.

**Stimuli.** We used the video footage of Experiment 1 as the stimulus material of Experiment 3. We decided to use the stimulus material from Experiment 1, instead of Experiment 2, as it was more natural and should facilitate imagining the hypothetical scenario required for Experiment 3, which in turn should help to increase the external validity of the findings.

**Measure and Procedure.** The procedure was identical to that of Experiment 1, except that this time participants were instructed with the following text: "Please imagine the following scenario. The referee has just made a decision against your team. You perceive this decision to be incorrect. Now you have to decide whether you argue with the referee or not." Participants responded to this scenario by logging in their answers on the digital scale "Would you argue with the referee in this situation" by moving the mouse cursor toward the left side of the scale (*not at all likely to argue*; i.e., value 0) or to the right side of the scale (*definitely argue*; i.e., value 100). This measure was based on previous prospective measures on the influence of NVB in guiding future behavior (Furley & Schweizer, 2014a). Again, we calculated our sample size to be able to detect small effects with a power of .8.

## Results

The mean "willingness to argue" rating was 43.16 ( $SD = 10.83$ ) for the ambiguous scenes and 40.17 ( $SD = 10.12$ ) for the unambiguous scenes. A paired-samples  $t$  test revealed a significant difference,  $t(57) = 2.424$ ,  $p = .01$ , one-tailed,  $d = 0.32$ , between the ratings for the unambiguous and the ambiguous scenes. Participants were significantly more likely to argue with the referee in the ambiguous scenes as compared with the unambiguous

scenes. Neither soccer experience nor the amount of soccer participants watched significantly influenced the pattern of results (i.e., these variables were not correlated with the difference in ratings for the ambiguous and unambiguous scenes).

## Discussion

Experiment 3 extended the leakage hypothesis by showing that participants anticipated being more likely to argue with a referee whose NVB was leaking a lack of confidence as he was communicating an ambiguous decision. This can be regarded as first evidence that the NVB of referees indeed has an influence on players. Although it is argued that NVB can be a facilitative tool referees can deliberately use, the results of the current study indicate that it can also out them when they are confronted with ambiguous decisions, potentially making it more likely for players to challenge these decisions. However, given the hypothetical nature of Experiment 3, future research is needed to investigate this theorizing in situ by testing whether and how (e.g., magnitude of behavioral reaction) athletes would in fact react to the NVB of referees.

## General Discussion

The aim of the current study was to investigate whether untrained observers detect differences in the NVB of soccer referees depending on whether they are communicating an unambiguous decision compared with an ambiguous decision. Experiments 1 and 2 showed that perceivers rated referees as less confident following ambiguous decisions and as more confident following unambiguous decisions. Occluding contextual information in Experiment 2 yielded results suggesting that it was indeed the referee's NVB that influenced observers' ratings and not the behavior of players that informed confidence ratings. Experiment 3 extended these findings by demonstrating that observers were influenced by a referee's NVB changes and anticipated being more likely to argue with the referee when his NVB signaled a lack of confidence.

Not only are the findings presented in line with evolutionary accounts of nonverbal expressions (Fridlund, 1994; Shariff & Tracy, 2011), the leakage hypothesis (Cozolino, 2006; Darwin, 1872/1988; Ekman, 2003), and the thin slices of behavior hypothesis (Ambady et al., 2000; Ambady & Rosenthal, 1992) but they add significantly to the growing literature on NVB in sports. Research on NVB in sports has shown that NVB of athletes occurring during the game can be interpreted as cues as to who is currently leading and who is trailing (Furley & Schweizer, 2014b), presumably because communicating status in confrontational encounters can be considered an adaptive mechanism organizing group life (Furley & Schweizer, 2016; Furley, Schnuerch, & Gibbons, 2016). Further research has shown that these NVB changes occurring during sports performance affect prospective confidence levels of opponent athletes

(Furley & Schweizer, 2014a). Similarly preperformance (Furley & Dicks, 2012; Furley, Dicks, & Memmert, 2012; Greenlees, Bradley, Thelwell, & Holder, 2005; Greenlees, Buscombe, Thelwell, Holder, & Rimmer, 2005) and post-performance NVB of athletes (Furley, Moll, & Memmert, 2015) has been shown to affect prospective confidence levels of both opponents and teammates and, indeed, can affect behavior and performance in sports competitions (Furley, Dicks, Stendtkke, & Memmert, 2012; Jordet & Hartman, 2008; Moll, Jordet, & Pepping, 2010). The present research extends these findings by showing that not only does an athlete's NVB change depending on the current situation but that this is also true for the referee. Importantly, these NVB changes have, in turn, the potential to affect subsequent social interaction, for the better or the worse, and have the potential of influencing the outcome of sport events.

The present results suggest that involuntarily communicating a lack of confidence in recent decisions might lead to negative consequences for referees. This suggestion is in line with training standards that emphasize the need for controlling referees' NVB (e.g., FIFA, 2015). In the following we will discuss how referees can gain more control over their NVB. One possibility would be to increase time and resources dedicated to training sessions on the control of NVB. The problem with this approach is that it is likely to fail precisely in situations when controlled NVB is needed most—in stressful situations after ambiguous incidents. This failure is predicted by several theoretical accounts, such as the leakage hypothesis (Ekman, 2003; Frank & Svetieva, 2013) and dual-process models of behavior (Furley, Schweizer, & Bertrams, 2015), and it is confirmed by our data. Another possibility would be to address the problem at an earlier level and to increase referees' confidence in their own decisions by training their decision-making abilities themselves. Theoretically, referees' lack of confidence should mainly leak when they are insecure regarding the correctness of their decision and when they lack capacity to control their NVB. Training programs that aim to improve referees' *intuitive* decision-making abilities should have the advantage of addressing both issues at once (Schweizer, Plessner, & Brand, 2013; Schweizer, Plessner, Kahlert, & Brand, 2011): Not only might they improve referees' decisions but they might also decrease the contribution of controlled processing to referees' decisions. Thereby, more cognitive capacity for controlling NVB might remain.

The present work has strengths but also some limitations. We consider it a strength that there were 25 stimuli in each experimental category, making it unlikely that our results were the product of only very specific stimuli (Wells & Windschitl, 1999). Still, it seems worthwhile to replicate our results using novel stimuli, maybe from different sports. Another strength is that the present work is firmly rooted in widely accepted theorizing (i.e., evolutionary theory, dual-process theory, and the thin slices of behavior approach), which makes the present results more likely to be robust and replicable (Wagenmakers,

Wetzels, Borsboom, van der Maas, & Kievit, 2012). Considering our design, our studies were adequately powered. Still, as always, replication attempts are well advised to use even larger samples. The main limitation is that although we attempted to rule out potentially confounding variables in Experiment 2, there might be confounding variables left in the stimulus material that pose alternative explanations for effects. However, the increase in effect size from Experiment 1 to Experiment 2 (in which the stimulus material was carefully modified to exclude other information besides the NVB of the referee) indicated that it was indeed the NVB of the referee that influenced confidence ratings.

A further limitation of the present research might be that we did not sample professional soccer players but had a sample of college students (with a major in sport science or physical education) with limited soccer experience. However, across all three experiments soccer experience did not influence the main finding, and there is no reason to believe that perceptions of referees' NVB might differ depending on domain-specific soccer knowledge (e.g., Furley & Schweizer, 2014b). In addition, participants' ratings were given in a laboratory context based on video-recorded behavior and were not sampled during actual match play, which might undermine the generalizability of the findings. Moreover, effect sizes were small to medium, with  $d$  values ranging from 0.33 to 0.67. Because the typical (average) effect in social psychology has been reported to be a  $d$  value of 0.41 (Richard, Bond, & Stokes-Zoota, 2003), we do consider the reported effects to be practically meaningful, especially in light of the relevance that is attributed to nonverbal communication by UEFA, FIFA, and referees themselves (Cunningham et al., 2014). Finally, the results of Experiment 3 require further empirical support to warrant the conclusion that referees' nonverbal displays of confidence affect the likelihood that athletes on the soccer field will argue or not. In real life this is likely to depend on more factors than just NVB of referees (e.g., exhaustion, their role in the team, and others). However, we think that NVB is one crucial factor among others. Taken together, we consider our results to be initial evidence that referees might be less able to control their NVB in stressful situations following difficult decisions and that this might lead to problematic consequences. Naturally, before refereeing associations start adapting their training routines, more research should be conducted.

In conclusion, the present research highlights that people are well equipped to pick up subtle changes in the NVB of referees that are associated with how confident the referees are in their decision. In line with the referees' belief that their communication has an important influence on player behavior (Cunningham et al., 2012), the present findings provide first experimental evidence for this belief. Importantly, the findings demonstrate that the wish of referees to project an "image of a competent, confident and decisive official" (Thatcher, 2005, p. 33) is complicated by a human's evolutionary inheritance of outing internal states such as a lack of confidence to observers.

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